Dimensional analysis of the spinless semiclassical Holstein Model

1 Adiabatic dimensionless dynamics

In the adiabatic limit, the electronic sector is assumed to relax instantaneously to the ground state corresponding to the instantaneous lattice configuration. Therefore, we do not explicitly evolve the electronic density matrix. Instead, for each lattice configuration $\{Q_i(t)\}$, the electronic Hamiltonian is diagonalized and the electronic occupation is evaluated in the instantaneous ground state. We start from the semiclassical Holstein Hamiltonian in the particle-hole symmetric form

$$H = H_e + H_{\text{lat}}, \tag{1}$$

where the electronic part is

$$H_e = -t_{\text{nn}} \sum_{\langle ij \rangle} \left(c_i^\dagger c_j + \text{h.c.} \right) - g \sum_i \left(n_i - \frac{1}{2} \right) Q_i, \tag{2}$$

and the classical lattice part is

$$H_{\text{lat}} = \sum_i \left[\frac{P_i^2}{2m} + \frac{K}{2} Q_i^2 \right]. \tag{3}$$

Here, Q_i and P_i are classical lattice displacement and momentum variables, while $n_i = c_i^\dagger c_i$ is the electronic density operator. The semiclassical equations of motion follow from Hamilton's equations for the lattice variables:

$$\dot{Q}_i = \frac{\partial H}{\partial P_i} = \frac{P_i}{m}, \tag{4}$$

$$\dot{P}_i = -\frac{\partial H}{\partial Q_i} = -K Q_i + g \left(\langle n_i \rangle - \frac{1}{2} \right), \tag{5}$$

where $\langle n_i \rangle$ is determined from the instantaneous electronic ground state. Equivalently,

$$m \ddot{Q}_i = -K Q_i + g \left(\langle n_i \rangle - \frac{1}{2} \right). \tag{6}$$

We introduce the characteristic lattice displacement scale

$$Q_0 = \frac{g}{K}, \tag{7}$$

and the corresponding momentum scale

$$P_0 = m \Omega Q_0, \qquad \Omega = \sqrt{\frac{K}{m}}. \tag{8}$$

The dimensionless variables are defined as

$$\tilde{Q}_i = \frac{Q_i}{Q_0}, \qquad \tilde{P}_i = \frac{P_i}{P_0}, \qquad \tilde{t} = \Omega t. \tag{9}$$

Thus the physical time is $t = \tilde{t}/\Omega$. Using these variables, the lattice equations become

$$\frac{d\tilde{Q}_i}{d\tilde{t}} = \tilde{P}_i, \tag{10}$$

$$\frac{d\tilde{P}_i}{d\tilde{t}} = -\tilde{Q}_i + \left(\langle n_i \rangle - \frac{1}{2}\right). \tag{11}$$

Equivalently, the second-order dimensionless equation of motion(also can be derived from Hellman-Feynman theorem) is

$$\frac{d^2\tilde{Q}_i}{d\tilde{t}^2} = \left(\langle n_i \rangle - \frac{1}{2}\right) - \tilde{Q}_i. \tag{12}$$

For a given lattice configuration, the electrons are governed by the dimensionless Hamiltonian

$$\tilde{H}_e[\{\tilde{Q}_i\}] = -\sum_{\langle ij \rangle} \left(c_i^\dagger c_j + \text{h.c.}\right) - 4\lambda \sum_i \left(n_i - \frac{1}{2}\right) \tilde{Q}_i, \tag{13}$$

where

$$\lambda = \frac{g^2}{KW}, \qquad W = 4t_{\text{nn}}. \tag{14}$$

At each time step, the instantaneous electronic ground state

$$|\text{GS}[\{\tilde{Q}_i\}]\rangle \tag{15}$$

is obtained from Eq. (13), and the local density is evaluated as

$$\langle n_i \rangle = \langle \text{GS}[\{\tilde{Q}_j\}] | n_i | \text{GS}[\{\tilde{Q}_j\}] \rangle. \tag{16}$$

This ground-state density is then inserted into Eq. (12) to update the lattice degrees of freedom. The adiabatic dynamics is a closed lattice-only evolution:

$$\boxed{\frac{d^2\tilde{Q}_i}{d\tilde{t}^2} = \left(\langle \text{GS}[\{\tilde{Q}_j\}] | n_i | \text{GS}[\{\tilde{Q}_j\}] \rangle - \frac{1}{2}\right) - \tilde{Q}_i}. \tag{17}$$

The electronic degrees of freedom enter only through the instantaneous ground-state expectation value $\langle n_i \rangle$.

Therefore, in the adiabatic dynamics, only relevant parameter is λ which is the interaction strength in terms of lattice oscillation timescale.

2 Thermal quench and dimensionless Langevin dynamics

We now include thermal effects phenomenologically by coupling the classical lattice variables to a Langevin bath with Markovian noise. In physical units, the stochastic equation of motion is

$$m\ddot{Q}_i = -KQ_i + g\left(\langle n_i \rangle - \frac{1}{2}\right) - \gamma \dot{Q}_i + \xi_i(t), \tag{18}$$

where γ is the damping coefficient and $\xi_i(t)$ is the stochastic force from the thermal bath. Using the dimensionless variables defined above, we obtain

$$\begin{aligned} \frac{d^2\tilde{Q}_i}{d\tilde{t}^2} &= -\tilde{Q}_i + \left(\langle n_i\rangle - \frac{1}{2}\right) - \frac{\gamma Q_0 \Omega}{g}\frac{d\tilde{Q}_i}{d\tilde{t}} + \frac{\xi_i(t)}{g} \\ &= -\tilde{Q}_i + \left(\langle n_i\rangle - \frac{1}{2}\right) - \tilde{\gamma}\frac{d\tilde{Q}_i}{d\tilde{t}} + \tilde{\xi}_i(\tilde{t}), \end{aligned} \tag{19}$$

Using $g = m\Omega^2 Q_0$ the dimensionless damping coefficient becomes

$$\tilde{\gamma} = \frac{\gamma}{m\Omega}. \tag{20}$$

with the dimensionless stochastic force as $\tilde{\xi}_i(\tilde{t}) = \frac{\xi_i(t)}{g}$. The thermal noise $\tilde{\xi}$ satisfies

$$\langle \tilde{\xi}_i(\tilde{t})\tilde{\xi}_j(\tilde{t}')\rangle = \frac{2\gamma k_B T_{\mathrm{bath}}}{g^2}\delta_{ij}\delta(t-t'). \tag{21}$$

Using $\delta(t-t') = \Omega\delta(\tilde{t}-\tilde{t}')$. we got,

$$\langle \tilde{\xi}_i(\tilde{t})\tilde{\xi}_j(\tilde{t}')\rangle = 2\tilde{\gamma}\tilde{T}\,\delta_{ij}\delta(\tilde{t}-\tilde{t}'), \tag{22}$$

where

$$\tilde{T} = \frac{K k_B T_{\mathrm{bath}}}{g^2} = \frac{k_B T_{\mathrm{bath}}}{\epsilon}. \tag{23}$$

with $\epsilon = \frac{g^2}{K}$ is the local polaron binding energy scale.

Thus, a thermal quench is characterized by the dimensionless bath temperature $\tilde{T}$, the dimensionless damping strength $\tilde{\gamma}$, and the electron–phonon coupling parameter λ appearing in the instantaneous electronic Hamiltonian. One caveat here is electronic dimensionless temperature for Fermi statistics is $\tilde{T}_{\mathrm{el}} = 4\lambda\tilde{T}$ since dimensionless electronic Hamiltonian (Eq. 13) is normalized by t_{nn}.

3 Simulation

To examine whether CDW order emerges dynamically, we performed a thermal-quench simulation of the dimensionless semiclassical Holstein model on a 20×20 square lattice at half filling. The parameters were chosen as $\lambda = 0.5$, $\tilde{\gamma} = 0.16$, and $\tilde{T}_{\mathrm{quench}} = 0.005$. The lattice degrees of freedom were evolved according to the Langevin equations

$$\frac{d\tilde{Q}_i}{d\tilde{t}} = \tilde{P}_i, \qquad \frac{d\tilde{P}_i}{d\tilde{t}} = \left(\langle n_i\rangle - \frac{1}{2}\right) - \tilde{Q}_i - \tilde{\gamma}\,\tilde{P}_i + \tilde{\xi}_i(\tilde{t}), \tag{24}$$

The initial state was prepared from a high-temperature($\tilde{T}_{\mathrm{init}} = 1.0$) Boltzmann distribution,

$$Q_i(0),\ P_i(0) \sim \mathcal{N}(0, T_{\mathrm{init}}), \tag{25}$$

and the system was then suddenly quenched to the low bath temperature $\tilde{T}_{\mathrm{quench}}$. For this moderate coupling strength, the system evolves toward a nearly uniform CDW configuration.

4 Dimensionless η damping

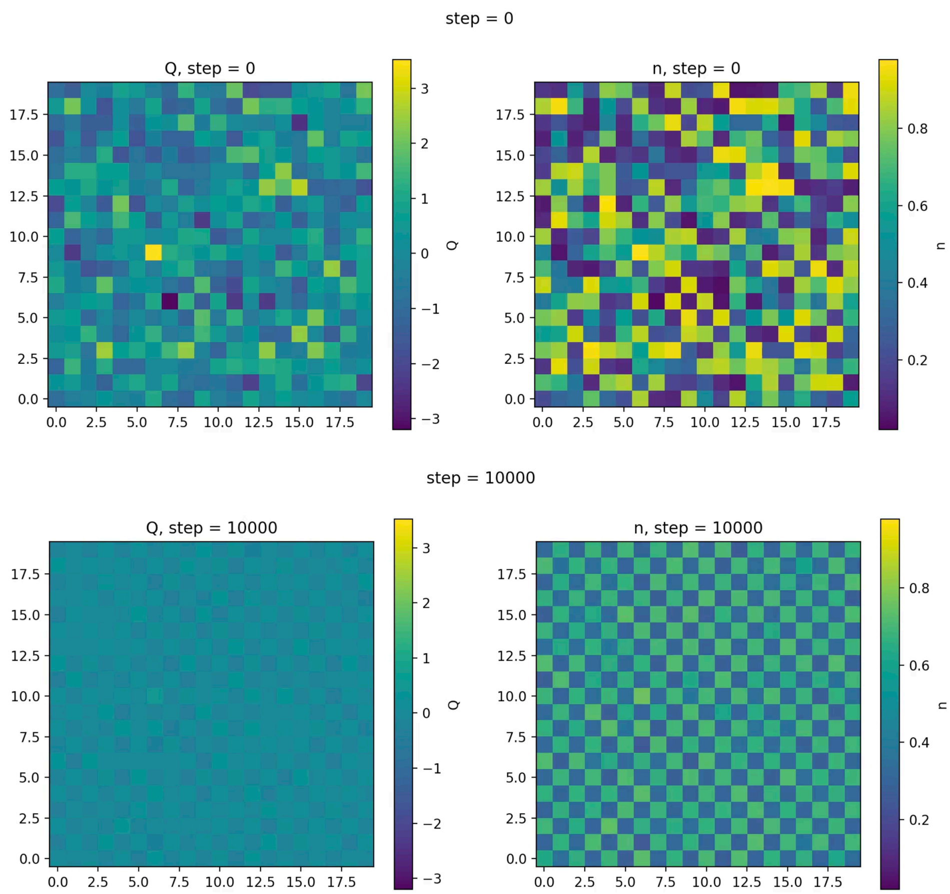

Figure 1: Snapshots of the real-space lattice configuration during the quench dynamics at iterations(steps) 0 and 10000. The initial state is drawn from a high-temperature random distribution with a large variance, while the late-time configuration develops a nearly uniform checkerboard pattern, indicating the emergence of CDW order after the quench.